

ISLAND LAKE, MB, Canada

WMO: 711450

Lat: 53.854N

Lon: 94.653W

Elev: 236

StdP: 98.53

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.2	-33.8	-40.6	0.1	-36.1	-38.3	0.1	-33.7	8.5	-18.3	7.7	-18.2	1.7	300	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	27.9	19.2	26.2	18.0	24.5	17.2	20.6	25.6	19.4	24.1	18.4	22.7	3.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	14.0	22.9	17.7	13.0	21.6	16.7	12.2	20.7	60.5	25.8	56.5	24.4	53.2	22.6	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.6	6.7	-39.3	31.9	2.4	1.9	-41.0	33.3	-42.4	34.4	-43.8	35.5	-45.5	36.8		
(5)			-39.4	22.7	2.3	1.6	-41.1	23.9	-42.5	24.8	-43.8	25.7	-45.5	26.9	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.1	-20.9	-18.5	-10.2	-0.9	7.3	14.9	18.4	17.0	11.1	3.3	-7.1	-17.1	(6)
(7)		DBStd	15.06	7.98	7.68	8.22	6.92	5.45	4.56	3.40	3.63	4.54	4.52	6.81	8.09	(7)
(8)		HDD10.0	4446	957	799	627	328	119	10	0	1	39	213	512	839	(8)
(9)		HDD18.3	6829	1215	1032	886	576	343	120	42	70	220	466	762	1098	(9)
(10)		CDD10.0	751	0	0	0	3	35	157	261	217	73	6	0	0	(10)
(11)		CDD18.3	93	0	0	0	0	1	17	44	28	3	0	0	0	(11)
(12)		CDH23.3	617	0	0	0	0	17	143	276	162	18	1	0	0	(12)
(13)		CDH26.7	116	0	0	0	0	2	28	53	31	2	0	0	0	(13)
(14)	Wind	WSAvg	3.2	2.9	2.8	3.0	3.0	3.1	2.9	2.9	3.2	3.4	3.9	3.7	3.0	(14)
(15)	Precipitation	PrecAvg	581	24	19	29	32	42	72	91	79	69	55	44	30	(15)
(16)		PrecMax	687	45	30	56	81	124	172	142	140	126	117	89	60	(16)
(17)		PrecMin	437	9	7	5	7	9	20	31	40	24	26	21	13	(17)
(18)		PrecStd	72	10	7	15	20	28	39	31	31	26	27	20	12	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.2	2.5	10.4	19.4	25.6	29.6	30.5	29.8	25.8	18.0	8.8	1.5	(19)
(20)			MCWB	-0.4	0.9	6.1	10.3	14.5	19.9	20.6	20.1	18.0	13.0	6.4	0.0	(20)
(21)		2%	DB	-2.6	0.2	6.6	15.4	22.3	26.7	28.0	27.1	22.6	14.5	5.7	-1.1	(21)
(22)			MCWB	-3.3	-0.6	2.8	8.1	12.9	17.6	19.4	19.2	16.6	10.5	3.5	-1.9	(22)
(23)		5%	DB	-5.6	-2.8	3.8	12.1	19.4	24.8	26.4	25.0	20.3	12.0	3.2	-3.0	(23)
(24)			MCWB	-6.2	-3.9	1.2	6.2	11.4	16.4	18.4	18.3	15.3	8.9	1.8	-3.5	(24)
(25)		10%	DB	-8.4	-6.5	1.4	9.1	16.6	22.8	24.6	23.0	18.0	9.7	1.4	-5.4	(25)
(26)			MCWB	-8.8	-7.4	-0.3	4.2	10.0	15.4	17.5	17.4	14.2	7.5	0.5	-5.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.2	1.2	6.4	10.9	15.8	21.7	22.3	21.6	19.5	13.8	7.1	0.1	(27)
(28)			MCDWB	0.4	2.0	9.7	18.4	23.1	28.3	28.1	26.4	23.8	16.6	8.8	1.1	(28)
(29)		2%	WB	-3.2	-0.7	3.0	8.4	13.8	19.2	20.8	20.2	17.6	11.3	3.7	-1.9	(29)
(30)			MCDWB	-2.6	0.2	5.8	14.7	19.8	24.5	25.8	25.2	21.1	13.8	5.4	-1.3	(30)
(31)		5%	WB	-6.1	-3.8	1.3	6.5	12.3	17.8	19.6	19.2	16.0	9.2	1.9	-3.5	(31)
(32)			MCDWB	-5.6	-2.9	3.5	11.6	17.8	22.4	24.2	23.6	19.4	11.5	3.1	-2.9	(32)
(33)		10%	WB	-8.8	-7.4	-0.1	4.6	10.7	16.5	18.5	18.1	14.4	7.7	0.5	-5.9	(33)
(34)			MCDWB	-8.3	-6.6	1.5	8.7	15.9	21.2	22.9	22.2	17.6	9.7	1.4	-5.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.8	11.3	12.3	11.8	10.9	10.0	9.1	8.3	7.4	5.3	5.8	7.6	(35)
(36)			MCDBR	9.4	10.5	12.1	14.8	15.3	12.8	11.4	11.0	10.9	9.1	6.5	7.5	(36)
(37)			MCWBR	9.2	9.5	9.1	9.1	7.8	5.7	4.9	5.3	6.0	5.9	5.2	7.1	(37)
(38)		5% WB	MCDWB	9.4	9.7	11.0	14.1	12.9	10.7	9.8	9.7	9.8	8.1	6.4	7.4	(38)
(39)			MCWBR	9.2	8.9	8.3	8.9	7.3	5.6	5.1	5.3	6.3	5.7	5.4	6.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.240	0.259	0.273	0.322	0.345	0.373	0.387	0.361	0.330	0.292	0.254	0.223	(40)	
(41)		taud	2.265	2.261	2.320	2.374	2.445	2.388	2.387	2.461	2.516	2.577	2.416	2.279	(41)	
(42)		Ebn at Noon	749	851	919	910	903	876	855	860	849	818	736	701	(42)	
(43)		Edh at Noon	57	81	96	106	105	112	111	97	81	61	50	46	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.92	2.02	3.46	5.11	5.39	5.68	5.46	4.46	3.06	1.66	0.85	0.64	(44)	
(45)		RadStd	0.09	0.13	0.25	0.40	0.47	0.47	0.37	0.37	0.37	0.19	0.08	0.09	(45)	

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page